Statistical Learning and Machine Learning

Exercises (Week 2)

Book exercises: 1.1, 1.12

Added Question: The random variables X and Y have joint pdf that is given by

$$f(x, y) = c\ sin(x + y)\ ,\ 0 \leq x, y \leq \pi/2$$

a) Find the value of the constant c

b) Find the joint cumulative distribution function (CDF) of X and Y

c) Find the marginal PDFs of X and Y i.e., $f(x)$ and $f(y)$

d) Find the mean, variance and covariance of X and Y

e) Find the conditional PDF $f_Y(y|x)$

f) Find the probability of the event: $P(Y > X \mid X = x),\ Hint:\ Use\ the\ result\ from\ part\ (e)$

Reading Exercise: Appendix C (Matrix Derivatives) from the course's textbook.

Statistical Learning and Machine Learning

Exercise Session (Week 2)

Classification and Loss Matrix

Measurements of an 1-dimensional continuous variable $x \in \mathbb{R}$ belonging to two classes (blue and red) follow the probability distributions illustrated in Figure 1. The class probability distributions are equal to $p(\mathcal{C}_1) = 2p(\mathcal{C}_2)$.

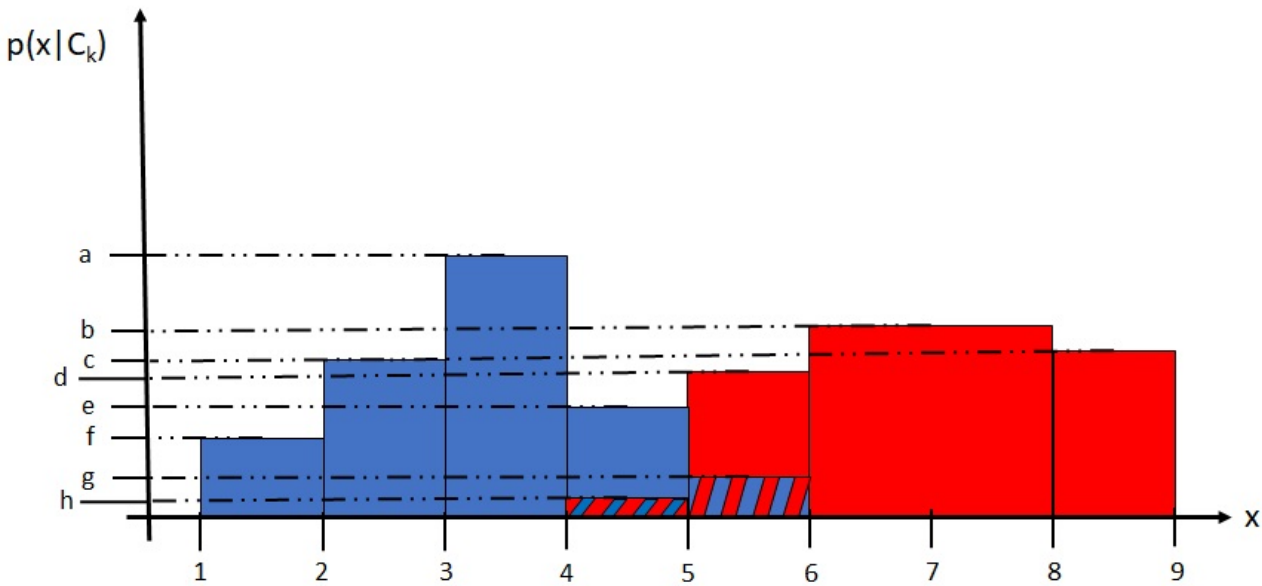

Fig. 1. Class distributions of measurements of an 1-dimensional variable x. Blue corresponds to $\mathcal{C}_1$ and red corresponds to $\mathcal{C}_2$. The values in the figure are equal to: $a = 0.42$, $b = 0.27$, $c = 0.22$, $d = 0.2$, $e = 0.17$, $f = 0.12$, $g = 0.07$, $h = 0.04$. Notice that for the values $4 < x \leq 5$ and $5 < x \leq 6$ the two classes are overlapping.

Because the importance of missclassifying measurements from the two classes is not the same, the following loss matrix is used:

$$\mathbf{L} = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \tag{1}$$

Answer the following questions:

1) What is the probability of observing a measurement with value of $x < 6$?
2) What is the probability of a measurement belonging to $\mathcal{C}_1$ to have a value of $x < 6$?
3) Describe a way to classify a measurement with value of $x = 4.5$ to one of the two classes using the probability and loss values provided in this exercise.